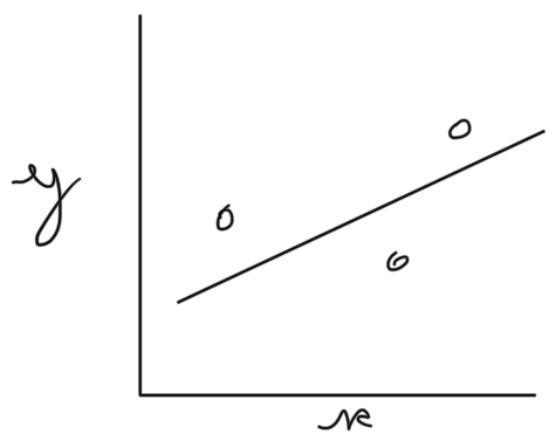




To find the best fitted line, we use gradient descent to find intercept only.

Let's assume slope to be 0.64

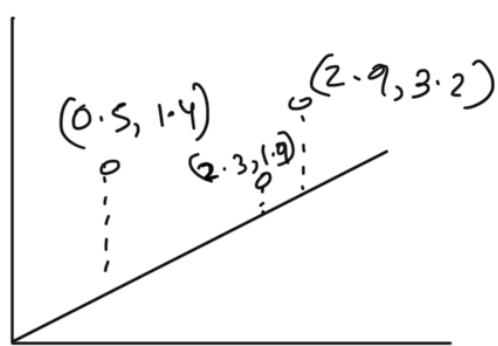
Later we use gradient descent to find slope and intercept as well.

$$h' = b + (a \times x)$$

$$\Rightarrow \text{Assume } h' = b + (0.64 \times x)$$

1.) Pick a random intercept to implement upon.

$$\text{let } b = 0$$

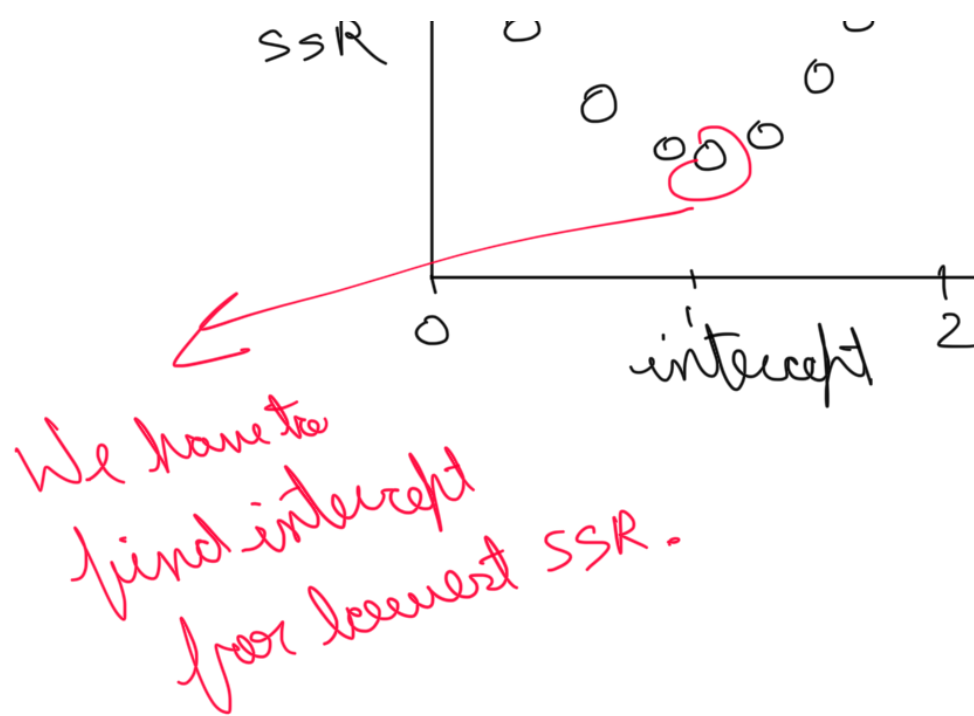


$$\therefore h' = 0.64 \times x$$

$$\text{Now, } SSR = 1.1^2 + 0.4^2 + 1.3^2 = 3.1$$

Now we plot SSR as intercept.



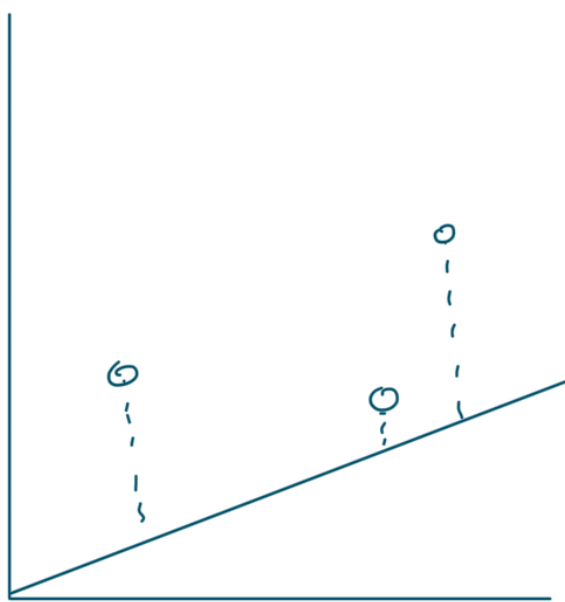


★ Now doing this repeatedly is tedious

Hence GD makes it simpler

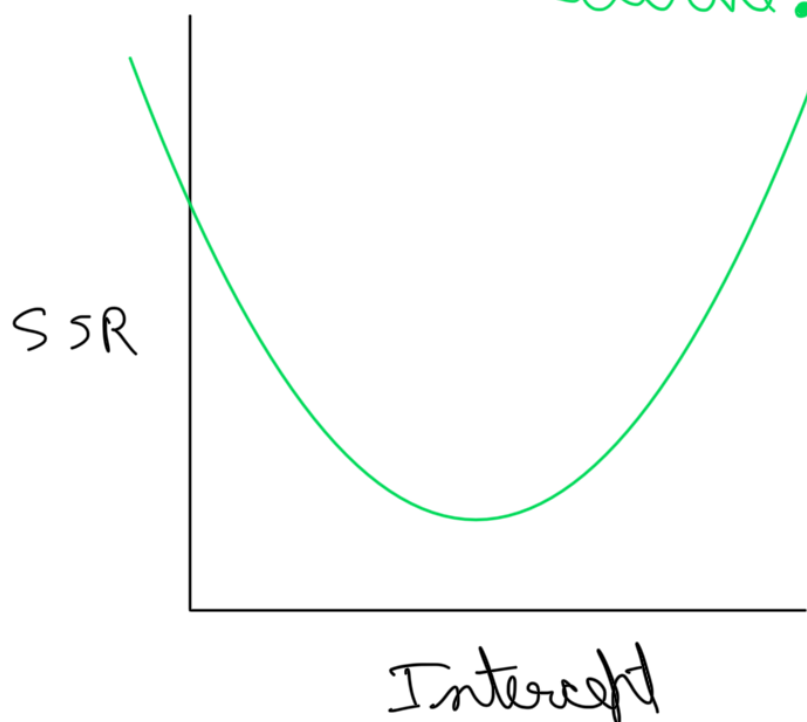


GD



$$SSR = (1.4 - (b + 0.64 \times 0.5))^2 + (1.9 - (b + 0.64 \times 2.3))^2 + (3.2 - (b + 0.64 \times 2.9))^2$$

we now have an equation for this curve.



⇒ Now we take derivative w.r.t. this curve w.r.t.

intercept.

$$\begin{aligned} SSR &= (1.4 - (b + 0.64 \times 0.5))^2 \\ &\quad + (1.9 - (b + 0.64 \times 2.3))^2 \\ &\quad + (3.2 - (b + 0.64 \times 2.9))^2 \end{aligned}$$

$$\begin{aligned} \frac{d(SSR)}{db} &= -2(1.4 - (b + 0.64 \times 0.5)) \\ &\quad - 2(1.9 - (b + 0.64 \times 2.3)) \\ &\quad - 2(3.2 - (b + 0.64 \times 2.9)) \end{aligned}$$

Note :- If we used Least Squares to solve for intercept, we would simply find where the slope of error = 0.

IN CONTRAST, GD finds min. value by taking steps from an initial guess until it reaches best value.

This is useful when $\frac{d(SSR)}{db} = 0$ is not possible in many cases.

Now, $d(SSR)$ - >

$$\frac{d}{dt} \int_{t=0}^{\dots} \dots$$

we take large steps
as we are far
from 0
(Not super huge)

Now in G.D,

$$\text{Step size} = -5.7 \times 0.1 = -0.57$$

Now calculate
new intercept.

$$\text{New intercept} = 0 - \text{Step Size}$$

$$= 0 - (-0.57) = 0.57$$

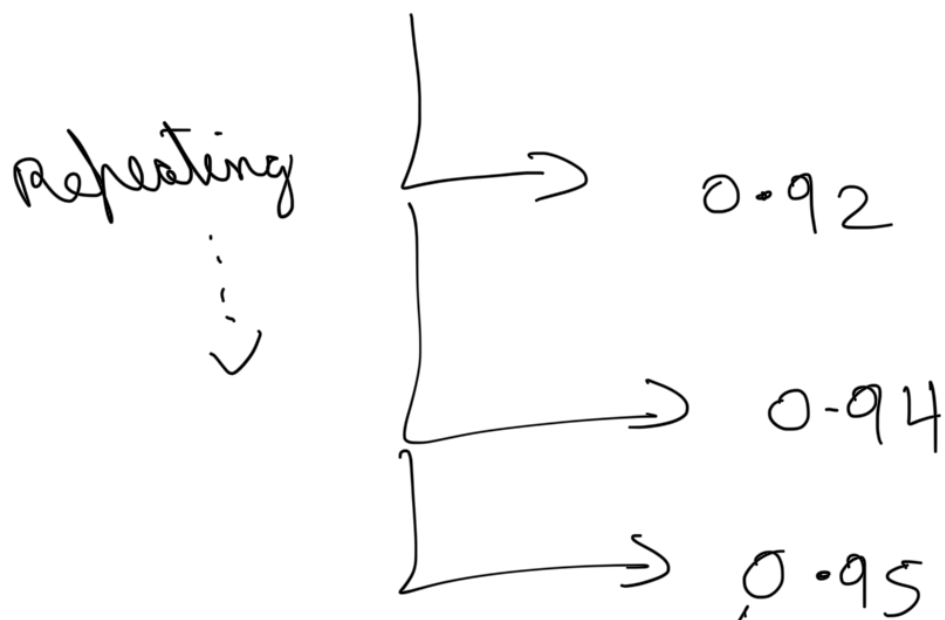
$$\text{Now } \frac{dSSR}{dt} \Big|_{t=0.57} = -2.3$$

$$\begin{aligned} \text{Step Size} &= -2.3 \times 0.1 \\ &= -0.22 \end{aligned}$$

$$\begin{aligned}\text{New intercept} &= 0.57 - (-0.23) \\ &= 0.57 + 0.23 \\ &= 0.8\end{aligned}$$

$$\frac{\Delta SSR}{\Delta b} \Big|_{b=0.8} = -0.9$$

$$\begin{aligned}\text{New } b &= 0.8 + 0.09 \\ &= 0.89\end{aligned}$$



After 6 steps, but

how to know, when to stop?

* In practice,

When step size = 0.001

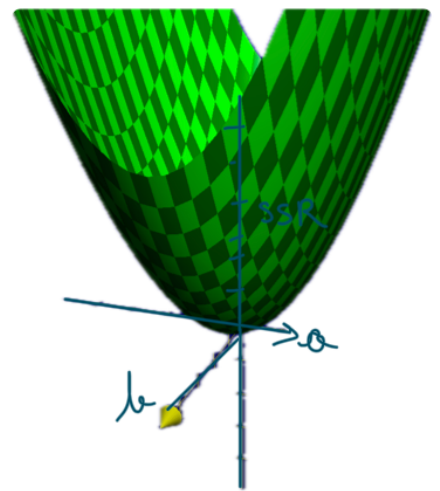
or
smaller

⇒ WE STOP

* OR steps = 1000

even if step size
is large.

Now this is the actual loss
function ^(SSR) dependant on slope
& intercept.



$$\begin{aligned} SSR = & (1.4 - (b + a \times 0.5))^2 \\ & + (1.9 - (b + a \times 2.3))^2 \\ & + (3.2 - (b + a \times 2.9))^2 \end{aligned}$$

Now we find $\frac{d SSR}{db}$ & $\frac{d SSR}{da}$

$$\frac{d(SSR)}{d\text{intercept}} =$$

$$\begin{aligned} \frac{d}{d \text{ intercept}} \text{ Sum of squared residuals} = & -2(1.4 - (\text{intercept} + \text{slope} \times 0.5)) \\ & + -2(1.9 - (\text{intercept} + \text{slope} \times 2.3)) \\ & + -2(3.2 - (\text{intercept} + \text{slope} \times 2.9)) \end{aligned}$$

&

$$\frac{d}{d\text{slope}} SSR =$$

$$\begin{aligned} \frac{d}{d \text{ slope}} \text{ Sum of squared residuals} = & -2(1.4 - (\text{intercept} + \text{slope} \times 0.5)) \\ & + -2(1.9 - (\text{intercept} + \text{slope} \times 2.3)) \\ & + -2(3.2 - (\text{intercept} + \text{slope} \times 2.9)) \end{aligned}$$

Here's the derivative of the Sum of the Squared Residuals with respect to the **Intercept**...



Just like before, we will start by picking a random number for the **Intercept**. In this case we'll set the **Intercept = 0...**

...and we'll pick a random number for the **Slope**. In this case we'll set the **Slope = 1.**

$$\begin{aligned} \frac{d}{d \text{ intercept}} \text{ Sum of squared residuals} = & -2(1.4 - (0 + 1 \times 0.5)) \\ & + -2(1.9 - (0 + 1 \times 2.3)) \\ & + -2(3.2 - (0 + 1 \times 2.9)) = -1.6 \end{aligned}$$

$$\text{Step Size}_{\text{Intercept}} = \text{Slope} \times \text{Learning Rate}$$

...now we plug the **Slopes** into the **Step Size** formulas...

$$\begin{aligned} \frac{d}{d \text{ slope}} \text{ Sum of squared residuals} = & -2 \times 0.5(1.4 - (0 + 1 \times 0.5)) \\ & + -2 \times 2.9(3.2 - (0 + 1 \times 2.9)) \\ & + -2 \times 2.3(1.9 - (0 + 1 \times 2.3)) = -0.8 \end{aligned}$$

$$\text{Step Size}_{\text{Slope}} = \text{Slope} \times \text{Learning Rate}$$

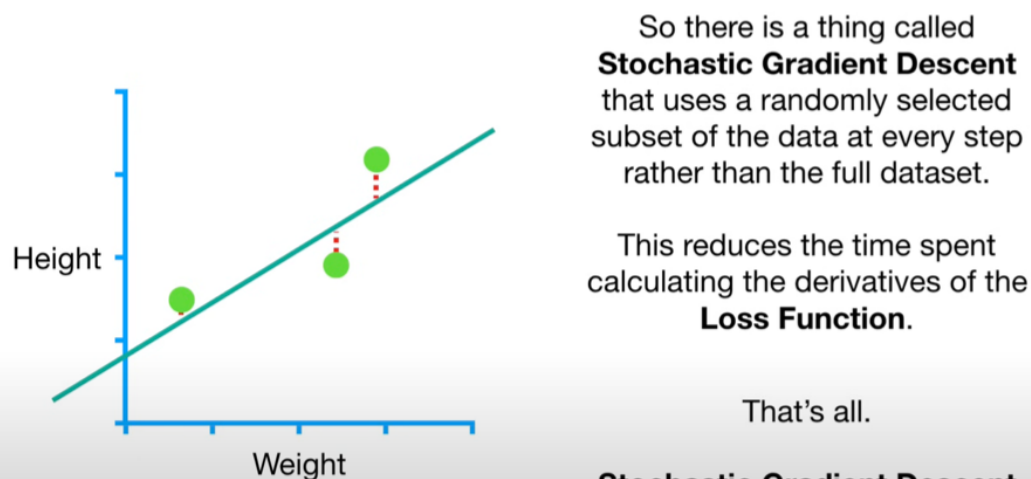
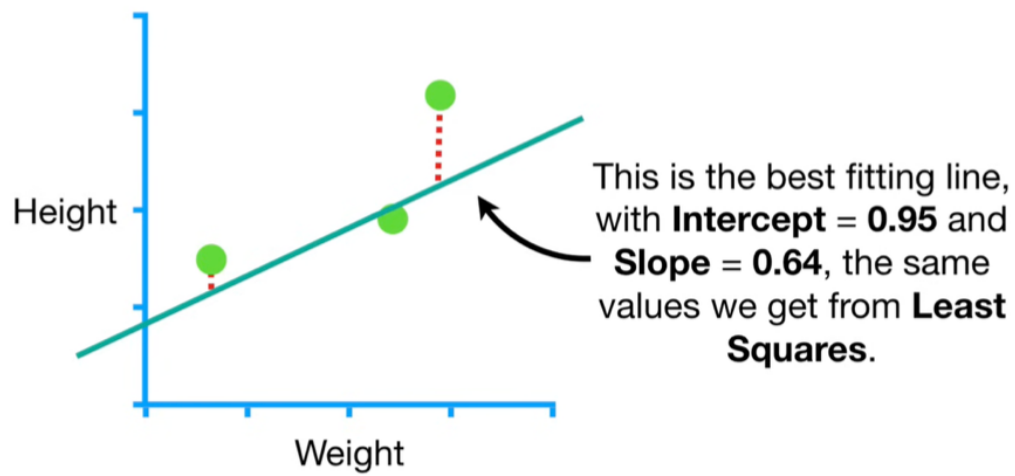
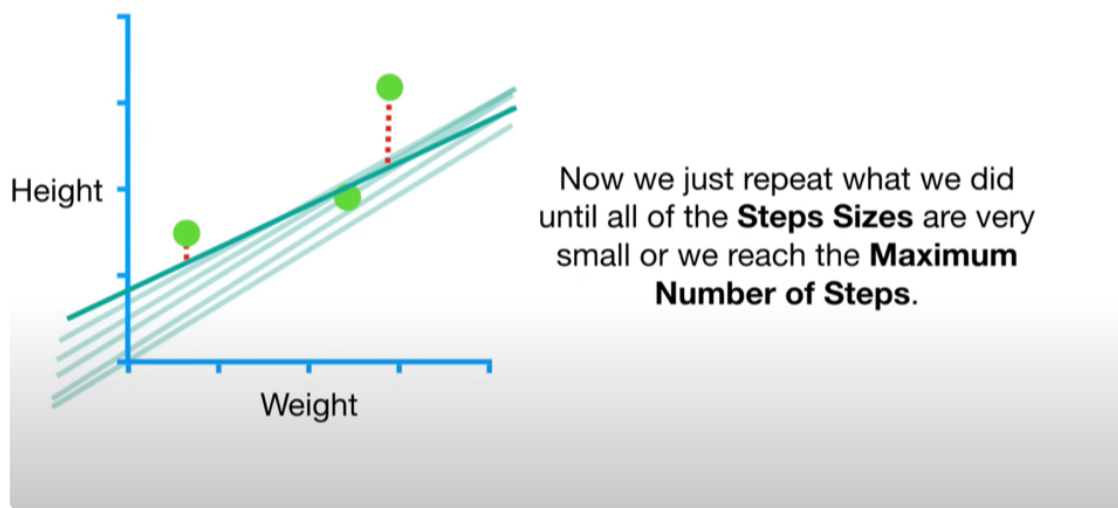
$$\begin{aligned} \frac{d}{d \text{ intercept}} \text{ Sum of squared residuals} = & -2(1.4 - (0 + 1 \times 0.5)) \\ & + -2(1.9 - (0 + 1 \times 2.3)) \\ & + -2(3.2 - (0 + 1 \times 2.9)) = -1.6 \end{aligned}$$

$$\text{Step Size}_{\text{Intercept}} = -1.6 \times 0.01$$

NOTE: The larger **Learning Rate** that we used in the first example doesn't work this time. Even after a bunch of steps, **Gradient Descent** doesn't arrive at the correct answer.

$$\begin{aligned} \frac{d}{d \text{ slope}} \text{ Sum of squared residuals} = & -2 \times 0.5(1.4 - (0 + 1 \times 0.5)) \\ & + -2 \times 2.9(3.2 - (0 + 1 \times 2.9)) \\ & + -2 \times 2.3(1.9 - (0 + 1 \times 2.3)) = -0.8 \end{aligned}$$

$$\text{Step Size}_{\text{Slope}} = -0.8 \times 0.01$$



Stochastic Gradient Descent sounds fancy, but it's no big deal.